



Build, Wiring, and Test Checklist

Celestial Electric Solar & Battery Room Monitor

Use this checklist while assembling the bench demo and before previewing the Arduino Project Hub page.

Planning

- ☐ Confirm the project is for low-voltage demonstration and training only.
- ☐ Confirm no Arduino connection will be made directly to line voltage, PV strings, ESS terminals, ATS controls, generator controls, fire alarm circuits, or service equipment.
- ☐ Choose BME280 as the primary temperature/humidity sensor, or use DHT20 as an alternate.
- ☐ Assign pin map before wiring and label each lead.

Bench wiring

- ☐ Mount Arduino on bench or in an enclosure with strain relief planned for any external leads.
- ☐ Connect the temperature/humidity sensor using I2C/Qwiic/STEMMA wiring.
- ☐ Connect the water leak sensor output to the assigned digital input.
- ☐ Connect the magnetic contact sensor using INPUT_PULLUP style wiring.
- ☐ Install red and green LEDs with 220 ohm current-limiting resistors.
- ☐ Use a multimeter to verify 5 V / 3.3 V rails and ground continuity.

Arduino Cloud setup

- ☐ Create a Thing for the Solar & Battery Room Monitor.
- ☐ Create dashboard variables for temperature, humidity, water leak, door/contact, normal status, and alert message.
- ☐ Associate the UNO R4 WiFi or selected Wi-Fi board with the Thing.
- ☐ Verify the device can connect to Wi-Fi and update cloud variables.

Functional test

- ☐ Confirm temperature and humidity display on the dashboard.
- ☐ Open and close the magnetic contact sensor; confirm dashboard state changes.
- ☐ Test water sensor with a controlled wet/dry test; dry the sensor after test.

- [] Confirm green LED is on when normal and red LED turns on during alert.
- [] Trigger each alert one at a time and confirm dashboard status is understandable.

Documentation

- [] Photograph the final bench setup and wiring.
- [] Save pin map, component list, and safety note with the project.
- [] Link the GitHub repository after code is uploaded.
- [] Preview Arduino Project Hub page before publishing.

Suggested pin map reference

Function	Recommended connection	Notes
BME280 / DHT20 I2C sensor	SDA/SCL or Qwiic/STEMMA connector	Use one temperature/humidity sensor. BME280 is the recommended primary option.
Water leak sensor digital output	D2	Verify wet/dry output state before final code. Invert logic if required.
Door/contact sensor	D3 and GND, INPUT_PULLUP	Closed contact to ground reads LOW. Open contact reads HIGH.
Green normal LED	D5 through 220 ohm resistor to GND	On when system is normal.
Red alert LED	D6 through 220 ohm resistor to GND	On when leak/contact/threshold alert is active.
USB-C power	USB-C port	Use USB power for bench demo. Use listed supply for any real installation.

Sign-off notes

Check	Initials / notes
Bench wiring verified	
Cloud dashboard updates verified	
Water leak input tested	
Door/contact input tested	
Normal and alert LEDs tested	
Safety note included in Project Hub description	
GitHub code/repository link checked	
Project preview reviewed before publish	

This project is for low-voltage monitoring, training, and demonstration only.

Do not connect an Arduino directly to service equipment, PV strings, battery terminals, inverter terminals, generator controls, transfer switches, utility equipment, fire alarm circuits, or any line-voltage wiring.